

Dan Hlevca

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U.S. Citizen — Eligible to obtain and maintain a Secret Security Clearance

SUMMARY

Experienced engineer and computational design practitioner with 10+ years of experience in programmatic CAD workflows, parametric geometry construction, and simulation-driven design automation. Built automated CAD pipelines using Siemens NX that programmatically construct, modify, and evaluate 3D assemblies for aerospace propulsion hardware. Deep experience with meshing strategies and simulation-in-the-loop design using ANSYS and OpenFOAM for iterative optimization. Developed Python-based automation scripts for geometry manipulation, batch processing, and design-space exploration. Strong background in CAD interoperability (STEP, IGES) and generating simulation-ready datasets from parametric models. M.S. in Data Science with ML/AI focus, combining computational design expertise with generative modeling and data-driven optimization. Ph.D. in Aerospace Engineering. U.S. Citizen.

TECHNICAL SKILLS

Programmatic CAD: Siemens NX (API scripting, automated geometry construction, parametric modeling), CAD interoperability (STEP, IGES), 3D assembly manipulation, batch geometry processing, automated model evaluation

Parametric Design & Optimization: Parametric component libraries, design-space exploration, geometry sweeps, shape optimization, simulation-in-the-loop design, topology-adjacent optimization methods

Meshing & Simulation: ANSYS Mesh, ANSYS Geometry, OpenFOAM, structured and unstructured mesh generation, mesh quality optimization, simulation-ready model preparation, CFD and thermal analysis

GenAI & Data-Driven Design: Machine learning for design optimization, reduced-order modeling (POD, SPOD, DMD), predictive modeling, generative design algorithms, data-driven modeling, Python-based automation

Programming: Python (NumPy, SciPy, Matplotlib), MATLAB, CAD API scripting, pipeline automation, scientific computing, batch processing workflows

Engineering Tools: Siemens NX, ANSYS Geometry, ANSYS Mesh, TecPlot, Git/GitHub, version-controlled development environments

AI-Native Workflow: Daily use of agentic coding tools (Codex, ChatGPT) for CAD scripting, simulation automation, data analysis, and engineering workflow optimization

EXPERIENCE

Senior Mechanical Engineer, Systems Thermal & Fluid Dynamics	June 2026 – Present
<i>CesiumAstro - Communication Systems for Space, Air and Ground</i>	<i>El Segundo, CA</i>

- Built automated CAD pipelines using Siemens NX that programmatically construct, modify, and evaluate 3D assemblies for regenerative cooling propulsion hardware
- Developed parametric geometry workflows enabling iterative design exploration and simulation-in-the-loop optimization of cooling channel configurations
- Created Python-based automation scripts for batch geometry processing, CAD interoperability (STEP), and simulation-ready model generation
- Applied meshing strategies and CFD analysis (ANSYS Fluent) to evaluate parametric design variants, feeding results back into geometry optimization loops
- Generated technical documentation for automated CAD workflows, parametric component libraries, and design exploration results

Lead Mechanical & Simulation Engineer	June 2024 – January 2025
<i>MATTUR - Powering a Better Future</i>	<i>Scottsdale, AZ</i>

- Built parametric CAD models and automated geometry workflows for iterative design exploration of shrouded wind turbine assemblies
- Performed simulation-in-the-loop design optimization: parametric geometry sweeps evaluated through CFD, with results feeding back into design modifications
- Applied linear and nonlinear optimization methods to geometry design, trading shape parameters against performance and packaging constraints

- Developed Python-based automation scripts for batch geometry processing, simulation execution, and post-processing of design variants
- Created simulation-ready meshed models using ANSYS Geometry and ANSYS Mesh for parametric design evaluation
- Validated simulation results against experimental data, calibrating models for reliable design-space exploration
- Generated technical documentation including parametric component specifications, design trade studies, and optimization reports

Senior Fluid Simulation Engineer

January 2023 – May 2024

SMOORE International

Scottsdale, AZ

- Built automated simulation pipelines using ANSYS Fluent and OpenFOAM for batch evaluation of parametric design variants
- Developed meshing workflows for complex 3D assemblies, optimizing mesh quality for simulation accuracy and throughput
- Applied reduced-order modeling techniques for rapid design-space exploration across geometric parameter sweeps
- Created Python-based automation scripts for geometry manipulation, simulation execution, and post-processing of results
- Designed experimental validation setups to calibrate simulation models for reliable parametric design evaluation
- Generated simulation-ready datasets from parametric CAD models, supporting data-driven design optimization
- Produced technical documentation for automated workflows, simulation methodologies, and design exploration results

Research Engineer - Numerical Aerodynamics and Fluid Mechanics

Nov. 2016 – July 2022

DynFluid Laboratory - Arts et Metiers ParisTech

Paris, France

- Developed automated simulation pipelines: meshing strategies, solver configuration, and post-processing workflows for parametric design evaluation of aerospace and automotive geometries
- Created numerical methods and mesh generation workflows for complex 3D geometries, optimizing mesh quality for simulation accuracy
- Built reduced-order models (POD, SPOD, DMD) for rapid prediction across geometric parameter sweeps, enabling high-throughput design exploration
- Applied simulation-in-the-loop design methodology: parametric geometry variants evaluated through CFD with results informing design modifications
- Produced peer-reviewed technical documentation, simulation methodologies, and design exploration reports for multi-year programs
- Collaborated with distributed international research teams, aligning on simulation standards and validation workflows
- Published peer-reviewed journal articles in AIAA, Experiments in Fluids, and European Journal of Mechanics/B Fluids
- Optimized sensor placement and measurement strategies for experimental data acquisition and simulation validation

Research Engineer - Experimental Aerodynamics and Fluid Mechanics

March 2014 – Oct 2016

Institut AeroTechnique

Saint Cyr l'Ecole, France

- Planned and executed experimental test campaigns: test matrix design, instrumentation strategy, sensor selection (thermocouples, pressure transducers, flow meters), and data acquisition
- Performed thermal-fluid testing using PIV, Kulite sensors, Pitot tubes, Kiel probes, and aerodynamic balance systems
- Coordinated implementation, qualification, and validation of measurement systems and instrumentation for experimental campaigns
- Correlated CFD simulations with experimental data, calibrating computational models against measured system performance
- Managed experimental data processing, statistical analysis, and data reduction into system performance models
- Identified and supported component-level and sub-system testing required for risk mitigation
- Developed test analysis reports and recommended design improvements based on experimental findings

Contractual Professor

September 2014 – Oct. 2016

EPF Engineering School

Sceaux, France

- Designed and delivered graduate-level curriculum in Numerical Methods in Fluid Mechanics for Aeronautics & Space master students, distilling complex CFD concepts for interdisciplinary audiences
- Created 2D and 3D geometry and mesh generation workflows using ANSYS, producing educational materials and technical documentation for student reference
- Developed problem-solving sessions, term projects, and assessments requiring students to articulate technical concepts through written reports and presentations
- Postprocessed simulation results with Python and TecPlot, creating data visualization materials for classroom instruction and knowledge transfer

Assistant Professor

September 2009 – March 2014

Technical University of Engineering

Bucharest, Romania

- Conducted professional development seminars for engineers on wind engineering and hydraulic machinery, distilling complex aerodynamic concepts for practicing professionals
- Designed and led experimental research programs, creating technical documentation and standard operating procedures for wind tunnel facilities
- Instructed senior-level fluid mechanics courses, developing assessments and providing detailed written feedback on student technical reports
- Lectured on laboratory techniques and theoretical framework, translating complex fluid mechanics principles into actionable knowledge for students
- Integrated wind tunnel instrumentation with data acquisition software, optimizing measurement workflows for experimental reliability

EDUCATION

M.S. in Data Science – ML/AI Focus

Tempe, Arizona

Arizona State University

Coursework: *Machine Learning, Artificial Intelligence, Data-Driven Modeling, Computational Methods for Engineering Systems, Applied Statistics*

Focus: *Application of ML/AI techniques to mechanical and aerospace systems, including fluid dynamics, control systems, and simulation-based optimization.*

Graduated: *May 2026*

Ph.D. in Aerospace Engineering – Aerodynamics & Flow Control

Bucharest, Romania

Technical University of Engineering | Joint French/Romanian European Research Grant

Sep. 2009 – Aug. 2012

Thesis: Numerical and experimental research on flow control in wind tunnels

B.S./M.S. in Mechanical and Aerospace Engineering

Sep. 2004 – Aug. 2009

Aerodynamics, Fluid Mechanics, Thermal Engineering

RESEARCH PROJECTS

FOSCO - Forcing Of Separation COntrol | *flow control, turbulent separated flows, pulsed jets, drag reduction.*

- The purpose of this project was to study the aerodynamic flow control at the rear of car vehicle, of interest to the automotive industry. Here we focused on obtaining reduced order models of the flow to help actuators to control the flow. The studied actuators are pulsating jets that completely destroy the recirculation bubble in the wake of the car.

CAJE – Contrôle Aérodynamique par JEts | *closed loop flow control, kulite sensors, reduced order modeling*

- The purpose of this project was to study the flow control at the rear of car vehicle, of interest to the automotive industry. This body reproduces a wake, characteristic of a large part of current automobile production. The objective here is to reduce aerodynamic drag, and therefore energy consumption. Through an experimental study in the wind tunnel, the idea is to test different active control strategies in order to understand their physical mechanisms, and to establish a number of recommendations for manufacturers wishing to deploy such systems. on their vehicles. Particular attention will be given to aerodynamic pulsating jets, by studying several model sizes.

CABOF – Contrôle Aérodynamique en Boucle Ouverte et Fermée | *numerical and experimental comparison*

- The objective here is to design and create a closed loop control strategy for flow around a car vehicle wake, characteristic of much of today's automotive production to reduce aerodynamic drag, and therefore fuel consumption. Through an experimental study in the wind tunnel, the idea is to test different control strategies in order to understand their physical mechanisms, and to establish a certain number of recommendations for manufacturers wishing to deploy such systems on their vehicles.
- Developed closed-loop aerodynamic flow control methodologies for aerodynamic performance enhancement and flow stability behavior

LANGUAGES

English: full professional proficiency
French: full professional proficiency
Spanish: limited professional proficiency
Romanian: native language

PUBLICATIONS & PEER REVIEW

1. **Modal analysis of actively controlled flow past a backward facing ramp - (2020) AIAA -**
<https://arc.aiaa.org/doi/10.2514/6.2020-0100>
2. **Atmospheric Boundary Layer Modeling in a Short Wind Tunnel - European Journal of Mechanics/B fluids (2020) -** <https://doi.org/10.1016/j.euromechflu.2019.10.003>
3. **Experimental study of the active control applied to the flow past a backward facing ramp - Experiments in Fluids (2018) -** <https://doi.org/10.1007/s00348-018-2497-0>
4. **Active Control applied to the flow past a backward facing ramp: S-PIV measurements and POD analysis - (2015) 3AF France -** <https://www.3af-aerodynamics.com>
5. **Atmospheric Boundary Layer Modeling in a Short Wind Tunnel - (2014) 3AF France -**
<https://www.3af-aerodynamics.com>
6. **Numerical predictions of the flow around a profiled casing equipped with passive flow control devices - (2013) Journal of Wind Engineering and Industrial Aerodynamics -**
<http://dx.doi.org/10.1016/j.jweia.2012.12.006>
7. **Penstock failure detection system at the “VÂLSAN” hydro power plant - (2012) - 26th IAHR Symposium on Hydraulic Machinery and Systems, Tsinghua University, Beijing, China**
8. **Device for passive flow control around vertical axis marine turbine - (2012) - 26th IAHR Symposium on Hydraulic Machinery and Systems, Tsinghua University, Beijing, China**

GRANTED PATENT

1. **Annular housing optimized for horizontal axis wind turbines -** <https://EuropeanPatentOffice.org>